05 · Data Binding

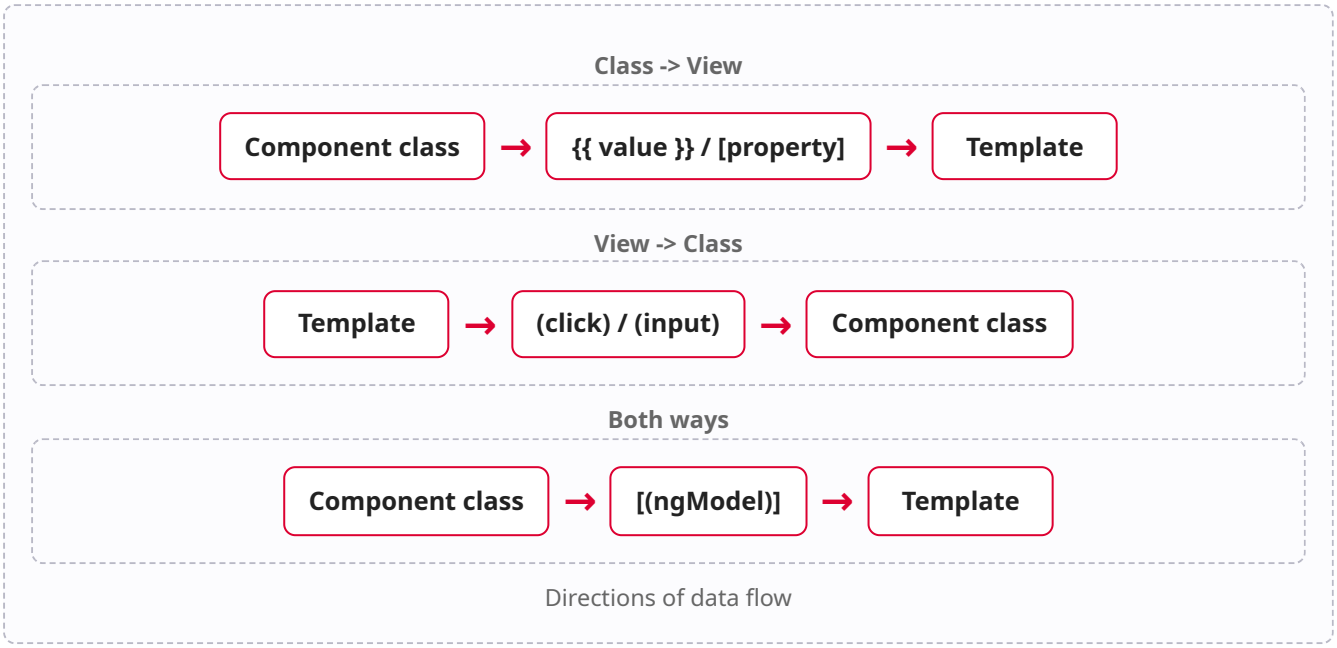
Beginner Angular learning notes

a. Definition

Data binding connects the component class to its template so the screen stays in sync with your data. Angular has four kinds: **interpolation** `{{ }}` (show a value), **property binding** `[prop]` (set an element property), **event binding** `(event)` (react to user actions) and **two-way binding** `[(ngModel)]` (both directions at once).

Remember the brackets: `[ ]` sends data *into* the template, `( )` sends events *out* of it, and `[( )]` (banana in a box) does both. For `ngModel` you must import `FormsModule`.

b. Diagram



Directions of data flow

c. Syntax

```
{{ expression }}                <!-- interpolation -->
<button [disabled]="isBusy">      <!-- property binding -->
<div [attr.aria-label]="label">   <!-- attribute binding -->
<button (click)="save()">         <!-- event binding -->
<input [(ngModel)]="name" />      <!-- two-way (needs FormsModule) -->
```

d. Example Program

`Programs/05_Data_Binding/data-binding.component.ts`

```
import { Component } from '@angular/core';
import { FormsModule } from '@angular/forms'; // needed for [(ngModel)]

@Component({
  selector: 'app-root',
  imports: [FormsModule],
```

```
  template: `
    <!-- interpolation: show a class value as text -->
    <h2>{{ brand }}</h2>

    <!-- two-way binding [(ngModel)] + attribute binding [attr.aria-label] -->
    <input [(ngModel)]="name" placeholder="Your name" [attr.aria-label]="'Name for ' + brand"
/>

    <!-- event binding (click) + property binding [disabled] -->
    <button (click)="greet()" [disabled]="name.trim() === ''">Greet</button>

    <p>You typed: {{ name }}</p>
    <p>{{ greeting }}</p>
    <p>Button clicked {{ clicks }} time(s)</p>
  `,
})
export class DataBindingComponent {
  brand = 'Chennai Tea House';
  name = '';
  greeting = '';
  clicks = 0;

  greet() {
    this.greeting = `Vanakkam, ${this.name}!`;
    this.clicks++;
  }
}
```

e. Output

Rendered in the browser (text on screen, after the UI steps below):

1. Type `Karthik` in the text box.
2. Click the **Greet** button twice.

```
Chennai Tea House
Greet
You typed: Karthik
Vanakkam, Karthik!
Button clicked 2 time(s)
```

f. Real-World Example (English + தமிழ்)

English

At a tea stall you tell the master 'less sugar' and read the price board. Information flows both ways: the board shows data to you, and your order goes back to the master.

தமிழ்

தேநீர்க் கடையில் நீங்கள் 'ஒரு டீ, சர்க்கரை குறைவா' என்று சொல்கிறீர்கள். மாஸ்டர் உடனே அதை கவனித்து அதன்படி டீ போடுகிறார்; விலைப்பட்டியல் பலகையில் உள்ள விலையை நீங்கள் படித்துக்கொள்கிறீர்கள். இரண்டு பக்கமும் தகவல் பரிமாறப்படுகிறது.

பலகையில் உள்ள விலை = `{{ price }}` (class-இலிருந்து திரைக்கு). நீங்கள் சொல்லும் ஆர்டர் = `(click)` அல்லது `(input)` (திரையிலிருந்து class-க்கு). இரண்டும் எப்போதும் ஒத்திசைவாக இருப்பது = `[(ngModel)]`.

g. Pros

- Less manual DOM code
- View always matches data
- Declarative and readable

h. Cons

- Wrong brackets cause confusing errors
- Heavy logic in templates hurts performance

i. Advanced / New Features

- `@let` template variables
- Signal-based `model()` for two-way binding
- Stricter template type checking

j. Common Mistakes

- Forgetting to import `FormsModule`, which causes "Can't bind to 'ngModel'".
- Writing `{{ }}` inside a property binding like `[title]="{{ x }}"`; use `[title]="x"` instead.
- Mixing up `[ ]` and `( )` brackets.
- Putting heavy logic or function calls inside templates; they run on every change detection.

k. Interview Questions

Q1. What are the types of data binding in Angular?
Interpolation, property binding, attribute binding, class/style binding, event binding and two-way binding.

Q2. Interpolation vs property binding?
Interpolation converts the value to a string inside text; property binding sets a DOM property with its real type (boolean, object...).

Q3. What is two-way binding?
Combining property and event binding, written `[(ngModel)]`, so the view and the class update each other.

Q4. When do you use `[attr.x]`?
When there is no DOM property for it (like `aria-*` or `colspan`), so you must set the HTML attribute.

l. Practice Exercises

1. Build a temperature converter: an input for Celsius and text showing Fahrenheit.
2. Add a button that is disabled until a checkbox is ticked.
3. Make a character counter that shows how many letters are typed in a textarea.